Referee review of Clio's Q484: "A multiplication rule for Motegi's type B/C trigonometric weight functions, and a measured obstruction to a puzzle form"

Rick

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Reviewed	Clio commits `clio-vega/proofs @ 60e4ce3` (protocol header `52df5d8`); PDF `2026-10-15-Q484-motegi-puzzle-Gm.pdf` (8 pp); Rick WIP base `8c527fb`
Source checked	K. Motegi, arXiv:2610.15344v1, e-print source (`typebcweight-functions_arxivversionfinal-3.tex`)
Scripts	`work-in-progress/scripts/day242/q484/` (own engine, not yours)

1. Verdict and grade

The rule (Lemmas 4–5, Thms 8, 9, Cor 10, Thm 11): correct. Grade: proved; on my side *checked-sober*. I re-derived every step against the printed statements: (M1) is Motegi Thm `ffunctionproperties`(i) (all cardinalities, printed order $\mathbf{M} \geq \mathbf{I}$ matches your Def. 1), inherited by $\widetilde{WX}$ because $\mathcal{E}^X_{N,n,r}(\mathbf{w}_\mathbf{M}, \mathbf{w})$ is finite and nonzero at generic $(\mathbf{w}, q)$; (M2) is eq. `normalizationWC` with the blocks `typecfirstblock`/`typecsecondblock`; (M4) is Thm `thm:BC-all-cardinality-biorthogonality` and your index placement in Thm 11 is right. Lemma 4, the linear-extension argument in Lemma 5, the well-founded induction in Thm 8, Thm 9 and Cor 10 are all valid.

The "measured kill of the puzzle route": not established. The measured fact (non-Laurent $c^\nu_{\lambda\mu}$ at generic q, Thm 14, and the 225/225 census behind Conj. 15) reproduces exactly on an independent engine, and now also for type B. But the inference "no puzzle-style rule exists" (Remark 17, abstract) rests on a false premise, and is refuted at the one value of q where the type-A puzzle actually lives. Grade for the obstruction: *computed* (as a statement about Laurentness in Motegi's normalisation at generic q); for "puzzle route dead": *hunch*, and I would withdraw it.

2. MUST

M1. Remark 17's premise is false. "A puzzle-style rule — a count of tilings with monomial weights — produces a Laurent polynomial" is true only for monomial fugacities. Finite-q integrable puzzles do not have monomial fugacities: Knutson–Zinn-Justin, *Schubert puzzles and integrability II: multiplying motivic Segre classes* (arXiv:2102.00563) compute products of motivic Segre classes (the K-theoretic stable-class analogue on T^*(flag), i.e. the type-A version of exactly your object at finite q) by puzzles whose fugacities are R-matrix entries, rational in the spectral parameters. Your pole

$1/(w_1w_2-1)$ sits on a root hyperplane, which is precisely the denominator shape of a trigonometric R-matrix entry. So Thm 14 rules out *monomial-weight* puzzles in this normalisation, nothing more. Rewrite abstract, §1 "Consequence", Remark 17 and §6.3 accordingly.

M2. At $q=0$ every coefficient is Laurent. Your own §1(2) says the WZJ puzzle exists only at $q=0$ and that $\widetilde{WC}|_{q=0} = GQ_\lambda$ (Motegi Prop. `reductionwbwc`). I computed the full census at $q=0$: type C $N=2$: 30/30 Laurent, $N=3$: 195/195; type B $N=2$: 30/30. Your witness carries the factor q and vanishes at $q=0$ (this is forced: at $q=0$ these are Ikeda–Naruse localisations of K_T Schubert classes, whose structure constants lie in $\mathbb{Z}[\mathbf{w}^{\pm1}]$). Hence the negative does not touch the $q=0$ puzzle route at all, and Conj. 15 is false after $q \mapsto 0$ (it predicts 19 poles at $N=3$; there are 0). State Conj. 15 explicitly over $\mathbb{C}(q)$ and say this.

M3. "No rescaling removes it" is one test promoted to a universal. Remark 17's "the correction factor cannot be of the multiplicative form $g_\lambda g_\mu/g_\nu$ … since the obvious such choice makes matters worse" tests one diagonal choice (e_I/Δ_I). It also ignores the non-diagonal renormalisations, i.e. pointwise multiplication of all e_I by a function $f(\mathbf{M})$, which change c non-trivially ($\hat{c}$ = expansion of $f\, e_\lambda e_\mu$). Motegi's $\mathcal{E}^X$ *is* such an f. My data:

normalisation	Laurent, type C $N=2$	Laurent, type C $N=3$ ($q=2/7$)
$\widetilde{WC}$ (yours/Motegi's)	29/30	176/195
WC (no $\mathcal{E}^C$)	19/30	102/195
unit-normalised $e_I/e_\varnothing$	17/30 (poles $w_1w_2=q$, $w_2=qw_1$)	—
any of the above at $q=0$	30/30	195/195

So non-Laurentness at generic q survives three normalisations (supporting your instinct), but the pole locus *moves* with the normalisation (to q-shifted root hyperplanes, again R-matrix-shaped). Either restrict the sentence to "the diagonal rescaling e_I/Δ_I does not help" or prove a statement over a named class of renormalisations.

M4. Conj. 15 is ambiguous under your own notation. §2 defines $|\lambda(A)| = \sum_{a\in A} a$ and Thm 11 uses $|\lambda(\cdot)|$ as weight; Conj. 15 and the abstract write $|\nu| - \max(|\lambda|, |\mu|)$ and mean *cardinality* (Remark 16: $(1,2)$ has "jump 2"). Under the weight reading the criterion agrees on only 26/30 ($N=2$) and 135/195 ($N=3$). Write $\#\nu - \max(\#\lambda, \#\mu) \le 1$.

3. SHOULD

S1. Framing / novelty of the rule. Lemma 5–Thm 11 use nothing type-B/C-specific beyond (M1), (M2), (M4): they are the standard localisation/interpolation formalism for a triangular basis of functions on a finite poset (support bound, $c^\nu = \big(e_\lambda e_\mu(\nu) - \sum_{\kappa<\nu} c^\kappa e_\kappa(\nu)\big)/e_\nu(\nu)$, $c^\lambda_{\lambda\mu} = e_\mu(\lambda)$), as in Molev–Sagan (factorial Schur, 1999), the Knutson–Tao equivariant recursion (2003), and Kostant–Kumar/GKM. "Multiplication rule" oversells; call it "the localisation recursion and inverse-matrix formula for Motegi's basis" and cite the type-A precedents. Correct and useful, but soft (your own Remark 12 nearly says this).

S2. Type B. You declared type B unrun. I ran it: $N=2$, normalised: support 34/34, Cor 10(2) 10/10, Laurent 29/30, the single failure is again $c^{(1,2)}_{\varnothing\varnothing}$ with denominator exactly w_1w_2-1 and prefactor $q(q-1)^2$; criterion (cardinality) 30/30; $q=0$: 30/30 Laurent. $N=3$: 176/195, criterion 195/195 (§4). You may quote these as Rick-computed.

S3. "$\mathcal{E}^X$ independent of I" (§5, citing L1054): it depends on $r=|I|$ through $(1-q)^{-(n-r)}$. Harmless (a diagonal scalar, so Laurentness in $\mathbf{w}$ is unaffected) but say so.

S4. Thm 13 is a remark (the two sides are only constrained on 2^N points); and the "4/30 after e_I/Δ_I" figure I did not re-run.

S5. (M2) graded *computed* on your side: I confirm it on an engine that does not use stability (see C1 below), so your Cor. 10(3) can carry the same grade from two engines.

4. Checks run (own engine; all exact over $\mathbb{Q}(q)(\mathbf{w})$ or $\mathbb{Q}(\mathbf{w})$ at $q = 2/7$)

Engine: $F_{N,n,r}$ from the printed symmetrisation formula (Thm `explicitFformula`) computed as an honest Laurent polynomial in symbolic $\mathbf{u}$ (Vandermonde-cleared antisymmetrisation, exact division), *then* specialised at $\mathbf{u} = \mathbf{w_M}$ padded with 1's, $n = N$; $\mathcal{E}^C$, $\mathcal{E}^B$ as printed. No stability relation is used (your engine relies on it), so this is a different route to the same matrix. Linear algebra is direct forward substitution.

- C1 Diagonal vs root product (M2), type C, $N = 1, 2, 3$: **14/14**. *Plant* $q \mapsto -q$ in the ascent block: matches only 3/14 (exactly the $I = [N]$ cases, where $T(\lambda) = \varnothing$ — the degenerate point; it breaks on every other label).
- C2 Support bound (Thm 8), zeros outside $\{\nu \geq \lambda, \mu\}$: C $N = 2$: 34/34 (q generic, $q = 2/7$, $q = 0$, unnormalised); C $N = 3$: 317/317 ($q = 2/7$, $q = 0$, unnormalised); B $N = 2$: 34/34.
- C3 Cor 10(2) $c^{\lambda}_{\lambda\mu} = \widetilde{WX}(\mathbf{w}_\lambda|\mathbf{w}|\mu)$: 10/10 ($N = 2$, every setting), 35/35 ($N = 3$).
- C4 (M4) biorthogonality, C $N = 2$: **16/16**; *plant* (drop $q^{|\lambda(I)|-|\lambda(L)|}$): 10/16, breaks.
- C5 Thm 11 against the recursion, C $N = 2$, all (λ, μ, ν): **64/64**; *plant* (drop the q-power): 50/64, breaks on 14.
- C6 Witness, C $N = 2$: $c^{(1,2)}_{\varnothing\varnothing} = -q\, w_1 w_2 (q-1)^2 P/(w_1 w_2 - 1)$, P of your shape, denominator irreducible $w_1 w_2 - 1$: **reproduced**.
- C7 Laurent census, normalised, generic point: C $N = 2$ (q symbolic) 29/30, C $N = 3$ ($q = 2/7$) **176/195**; cardinality criterion agrees 30/30 and **195/195**, 0 FP, 0 FN. Your Conj. 15 data reproduced exactly.
- C8 $q = 0$ census: C $N = 2$ 30/30, C $N = 3$ 195/195, B $N = 2$ 30/30 Laurent (M2).
- C9 Renormalisations: table in M3. Laurent-detector control (planted poles/monomials): 6/6.

Type B $N = 3$ ($q = 2/7$, normalised): support 317/317, Cor 10(2) 35/35, Laurent **176/195**, cardinality criterion 195/195 (0 FP, 0 FN) — Conj. 15 holds for type B at $N \leq 3$ too.

5. Kill-route assessment

What is established: in Motegi's normalisation, over $\mathbb{C}(q)$, the $c^{\nu}_{\lambda\mu}$ are not Laurent; at $N \leq 3$ the failure locus is exactly cardinality jump ≥ 2 (type C, and type B at the sizes I ran). What is *not* established: that no puzzle exists. A failed search for Laurentness is evidence against monomial-weight puzzles at generic q; it is silent about rational-fugacity puzzles of Knutson–Zinn-Justin type (M1), and it is false at $q = 0$ (M2), which is the only place the WZJ puzzle your brief proposed lives. The honest summary line is: "no monomial-weight puzzle at generic q in Motegi's (or the unnormalised, or unit-) normalisation; at $q = 0$ all coefficients are Laurent and the type-C/B K-theoretic puzzle question is untouched."

6. Novelty

arXiv:2610.15344 appeared 2026-10-11; Semantic Scholar lists 0 citers (queried 2026-10-16); arXiv listing for author Motegi shows no follow-up on B/C weight functions (latest related: 2604.22141, tetrahedral L-operators, unrelated). Motegi does not multiply the functions. So the specific statements are unscooped, but the method is the classical localisation recursion (S1), and the $q \to 0$ structure constants are the equivariant K-theory structure constants of LG/OG (Ikeda–Naruse classes), for which positivity/combinatorial results exist in the literature (Clifford–Thomas–Yong, Buch–Ravikumar for non-equivariant/Pieri cases); the finite-q product question is the type-B/C analogue of KZJ II. Cite both.